INCH-POUND
A-A-59440/1D
02 July 2020
SUPERSEDING
A-A-59440/1C
06 December 2018

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

COCK-VALVE, CENTER DRAIN, INTERNAL SEAT OR EXTERNAL SEAT, 150 PSI, BRASS, NON-FLAMMABLE LIQUID OR GAS, NPTF-1 THREADS

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

SCOPE. The complete requirements for procuring the cock-valve, center drain, internal seat or external seat, 150 psi, brass, non-flammable liquid or gas, NPTF-1 threads described herein shall consist of this document and the issue in effect of A-A-59440.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the PIN as shown in the following example (see NOTES).

AA59440	/01	-	001	J
CID number	Applicable CID spec sheet (two digits)		Dash number (see figure 1 and figure 2)	Handle plating (see Table I)

SALIENT CHARACTERISTICS.

Interface and physical dimensions. Cock-valves, center drain, internal seat or external seat, 150 psi, brass, non-flammable liquid or gas, NPTF-1 threads supplied to this CID specification sheet shall be as specified herein (see figure 1 and figure 2).

Sizes. Nominal sizes of 1/4 inch through 1 inch with NPTF-1 threads are covered by this CID.

Intended applications. These items are for use in non-flammable liquid or gas systems where pressures do not exceed 150 psig (10.3 bar) at 150°F (65.6°C) and the liquid or gas is compatible with brass. Not for use in potable water systems.

Permissible variations in dimensions. Figures 1 and 2 lists the permissible variation in dimensions for each size. The external dimensions of the cock-valve should not limit procurement unless the end item applications restrict these dimensions.

Optional designs. Figures 1 and 2 are for identification purposes only and are not intended to restrict designs or shapes not dimensioned.

Valve bodies. Valve bodies shall be provided with hexagonal wrenching surfaces.

AMSC N/A

FSC 4820

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Threads. Threads are dryseal NPTF-1 in accordance with ASME B1.20.3 per A-A-59440.

Material.

Brass. Body and handle, external seat only, shall be brass in accordance with the following commercial standards:

ASTM B16/B16M	ASTM B121/B121M	ASTM B505/B505M	SAE AMS4845
ASTM B22	ASTM B124/B124M	ASTM B584	SAE AMS4855
ASTM B30	ASTM B148	ASTM B763	SAE AMS4860
ASTM B36/B36M	ASTM B176	ASTM B770	SAE AMS4862
ASTM B61	ASTM B271/B271M		SAE AMS4890
ASTM B62	ASTM B369		

Carbon steel. Cast, forged, or machined carbon steel shall be in accordance with QQ-S-698.

Handle internal seat (see figure 1). Handle may be brass, hot rolled or cold rolled steel conforming to QQ-S-698 or another commonly accepted industry standard.

Handle external seat (see figure 2). Handle may be hot rolled or cold rolled steel conforming to QQ-S-698 or another commonly accepted industry standard.

Protective coating for handle:

TABLE I. Material and finish identification codes.

PIN code material/plating finish	Material	Plating finish
Blank	Steel or Brass	Brass with no plating or steel with Zinc Plating. (See Note 6/ and Table II) 3/ 7/
B	Brass	No additional plating. 3/
C	Carbon Steel	Cadmium in accordance with SAE-AMS-QQ-P-416, type II, class 2. 1/ 2/ 6/
Z	Carbon Steel	Zinc in accordance with ASTM B633; type V or VI, Fe/Zn 8. 5/
J	Carbon Steel	Zinc-nickel in accordance with SAE-AMS2417, 12% to 16% nickel, type 2, grade B. (315 ¼ µ inches (8¼ µm) minimum) 4/
H	Carbon Steel	Zinc-Aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 4/ 6/

1/ Embrittlement test need not be run.
2/ Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA-owned shipboard systems, unless otherwise approved by NAVSEA. The use of cadmium plating for Army ground vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement. Pin code J may be substituted for cadmium (C PIN code) for Army ground vehicle applications.
3/ Brass with lead contents shall not be used in potable water systems.
4/ Hexavalent chromium free. Finish shall be RoHS compliant.
5/ For use in low corrosion environments.
6/ For use in medium to high corrosion environments.
7/ These parts were originally listed in MS35782 as a single part type.

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NOTE: The following restrictions do not apply to manufactures of the cock valve and is applicable only to OEM's and users.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

The use of cadmium plating or coatings depositing hexavalent chromium for Army vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement.

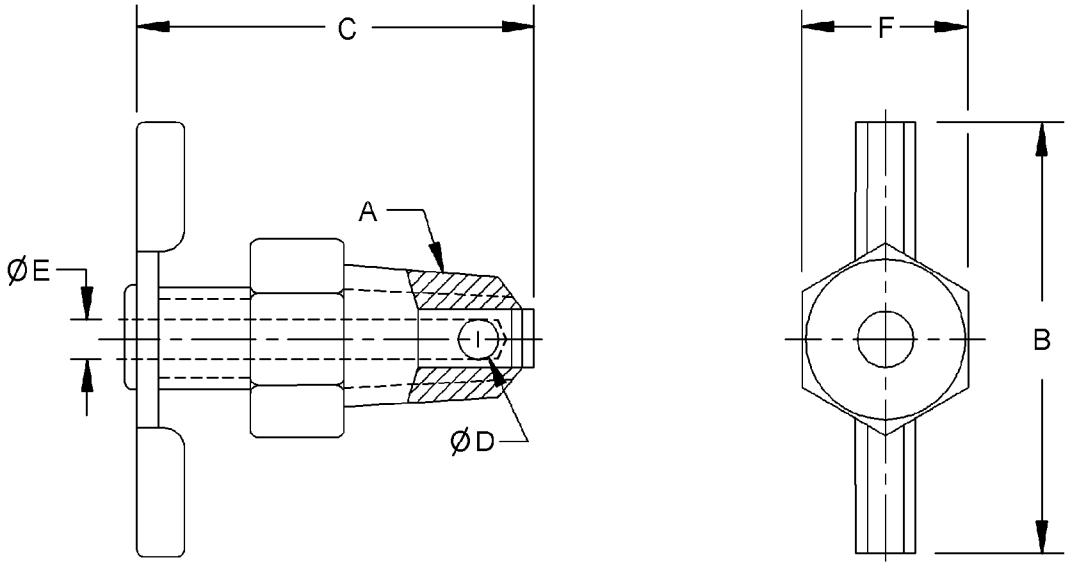
For new designs PIN codes C and Z are not to be supplied to Army and Marine ground forces.

Class I and II ozone depleting substances (ODS) shall not be used in A-A-59440/1 or any referenced procedures.

NOTE: End of restrictions only applying to OEM's and users.

Test pressures:

a. Shell: 150 psig (10.3 bar).
b. Seat: 50 psig (3.45 bar).



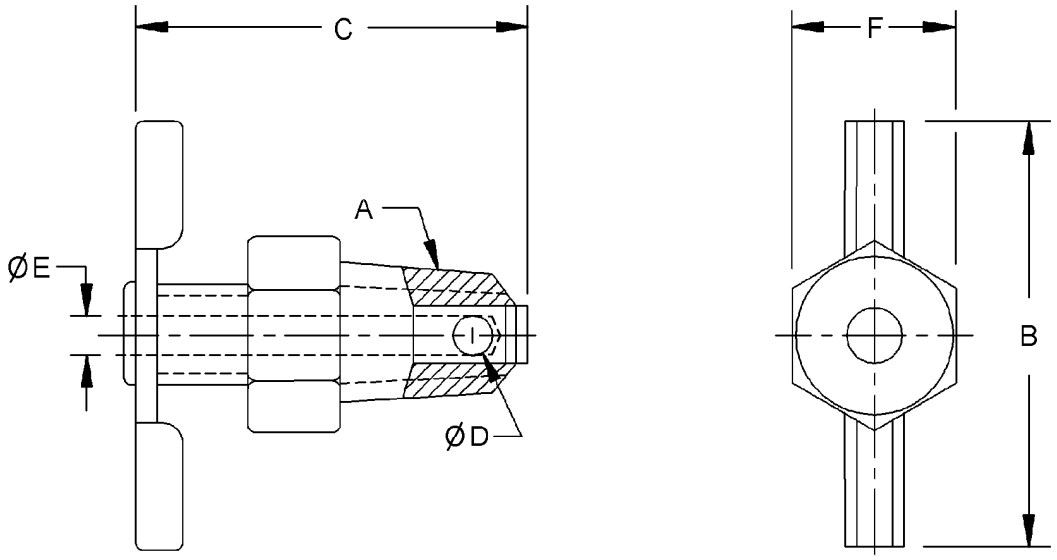
Dash number	Handle plating	A Pipe Thread NPTF-1	B maximum (mm)	C open maximum (mm)	D diameter minimum (mm)	E diameter minimum (mm)	F ±.015 (0.38 mm)
001	Zinc or Brass	1/8-27	1.38 (34.0)	1.88 (47.8)	.19 (4.8)	.12 (3.1)	.44 or .56 (11.2 or 14.2)
001B	Brass						
001C	Cadmium						
001Z	Zinc						
001J	Zinc-nickel						
001H	Zinc-Aluminum						

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002	Zinc or Brass	1/4-18	.50 (38.1)	2.00 (50.8)	.19 (4.8)	.19 (4.8)	.56 or .62 (14.2 or 15.7)
002B	Brass						
002C	Cadmium						
002Z	Zinc						
002J	Zinc-nickel						
002H	Zinc-Aluminum						
003	Zinc or Brass	3/8-18	1.50 (38.1)	2.00 (50.8)	.31 ±.02 (7.6 ± 0.5)	.25 to .31 (6.4 to7.9)	.69 or .75 (17.5 or 19.1)
003B	Brass						
003C	Cadmium						
003Z	Zinc						
003J	Zinc-nickel						
003H	Zinc-Aluminum						

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.010 inch (0.25 mm).
4. Diameter of hole shall be adequate to allow drainage through the valve.

FIGURE 1. Dimensions and configuration internal seat.



Dash number	Handle plating	A Pipe Thread NPTF-1	B maximum (mm)	C closed reference (mm)	D diameter reference (mm)	E diameter minimum (mm)	F ±.015 (0.38 mm)
004	Zinc or Brass	1/8-27	1.31 (33.3)	.90 (22.86)	.09 (2.3)	.12 (3.0)	.44 or .50 (11.2 or 12.7)
004B	Brass						
004C	Cadmium						
004Z	Zinc						
004J	Zinc-nickel						
004H	Zinc-Aluminum						
005	Zinc or Brass	1/4-18	1.38 (35.1)	1.00 (25.4)	.16 (4.1)	.19 (4.8)	.50 or .56 (12.7 or 14.2)
005B	Brass						
005C	Cadmium						
005Z	Zinc						

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005J	Zinc-nickel						
005H	Zinc-Aluminum						
006	Zinc or Brass	3/8-18	1.6 (42.93)	1.44 (36.6)	.16 (4.1)	.19 (4.8)	.62 or .69 (15.7 or 17.5)
006B	Brass						
006C	Cadmium						
006Z	Zinc						
006J	Zinc-nickel						
006H	Zinc-Aluminum						
007	Zinc or Brass	1/2-14	2.00 (50.8)	1.62 (41.1)	.31 (7.9)	.38 (9.7)	.81 or .88 (20.6 or 22.4)
007B	Brass						
007C	Cadmium						
007Z	Zinc						
007J	Zinc-nickel						
007H	Zinc-Aluminum						
008	Zinc or Brass	1-11 1/2	2.50 (63.5)	2.31 (58.7)	.53 (13.5)	.75 (19.1)	1.38 or 1.50 (35.1 or 38.1)
008B	Brass						
008C	Cadmium						
008Z	Zinc						
008J	Zinc-nickel						
008H	Zinc-Aluminum						

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.010 inch (0.25 mm).
4. Dimension “C” open/closed length dimensions may vary from manufacturer to manufacturer.
5. Diameter of hole shall be adequate to allow drainage through the valve.

FIGURE 2. Dimensions and configuration external seat.

NOTES:

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

FEDERAL SPECIFICATION

QQ-S-698 - Steel, Sheet and Strip, Low Carbon

COMMERCIAL ITEM DESCRIPTION

A-A-59440 - Cock-Valves and Valves, General Requirements for

(Copies of these documents are available online at https://quicksearch.dla.mil or https://assist.dla.mil).

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Other Publications

ASME INTERNATIONAL

ASME B1.20.3 - Dryseal Pipe Threads (Inch)

(Copies of these documents are available online at https://www.asme.org)

ASTM INTERNATIONAL

ASTM B16/B16M	-	Free-Cutting Brass Rod, Bar and Shapes for use in Screw Machines
ASTM B22	-	Bronze Castings for Bridges and Turntables
ASTM B30	-	Copper Alloys in Ingot Form
ASTM B36/B36M	-	Brass Plate, Sheet, Strip, and Rolled Bar
ASTM B61	-	Steam or Valve Bronze Castings
ASTM B62	-	Composition Bronze or Ounce Metal Castings
ASTM B121/B121M	-	Leaded Brass Plate, Sheet, Strip, and Rolled Bar
ASTM B124/B124M	-	Copper and Copper Alloy Forging Rod, Bar, and Shapes
ASTM B148	-	Aluminum-Bronze Sand Castings
ASTM B176	-	Copper-Alloy Die Castings
ASTM B271/B271M	-	Copper-Base Alloy Centrifugal Castings
ASTM B369	-	Copper-Nickel Alloy Castings
ASTM B505/B505M	-	Copper Alloy Continuous Castings
ASTM B584	-	Copper Alloy Sand Castings for General Applications
ASTM B633	-	Electrodeposited Coatings of Zinc on Iron and Steel
ASTM B763	-	Copper Alloy Sand Castings for Valve Applications
ASTM B770	-	Copper-Beryllium Alloy Sand Castings for General Applications
ASTM F1136/F1136M	-	Zinc/Aluminum Corrosion Protective Coatings for Fasteners

(Copies of these documents are available online at http://www.astm.org or from the ASTM International, P.O. Box C700, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

SAE-AMS-QQ-P-416	-	Plating, Cadmium (Electrodeposited)
SAE AMS4845	-	Tin Bronze Castings, Sand and Centrifugal 87.5CU - 10SN - 2ZN as Cast
SAE AMS4855	-	Leaded Red Brass, Sand and Centrifugal Castings 85CU - 5.0SN - 5.0PB - 5.0ZN as Cast
SAE AMS4860	-	Manganese Bronze, Sand and Centrifugal Castings 58CU - 39ZN - 1.2FE - 1.0AL - 0.80MN as Cast
SAE AMS4862	-	Manganese Bronze, Sand and Centrifugal Castings 63CU - 24ZN - 6.2AL - 3.8MN - 3.0FE High Strength, as Cast
SAE AMS4890	-	Copper-Beryllium Alloy Castings 97CU - 2.1BE - 0.52CO - 0.28SI Solution Heat Treated (TBOO)
SAE-AMS2417	-	Plating, Zinc-Nickel Alloy

(Copies of these documents are available on line at www.sae.org from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, and Tel: 877-606-7323 [inside USA and Canada] or 724-776-4970 [outside USA], email at CustomerService@sae.org.)

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Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

MFR's CAGE	MFR's name and address
01276	Eaton Aeroquip Inc. 14615 Lone Oak Rd. Eden Prairie, MN 55344 952-294-2032/888-258-0222 jeffenoe@eaton.com
93061	Parker-Hannifin Corp Brass Products Division 300 Parker Drive Otsego, MI 49708-1472 260-636-2104 dwsanders@Parker.com
20969	Essex Brass Corp 23500 Pinewood Warren, MI 48091-3122 586-757-8200 bpushman@essexbrass.com

Military standard (MS) supersession data. This CID replaces the following MS as shown in table II.

TABLE II. MS supersession data.

CID PIN 1/	Superseded MS
AA59440/1-001	MS35782-1
AA59440/1-001B	MS35782-1
AA59440/1-001C	MS35782-1
AA59440/1-001Z	MS35782-1
AA59440/1-002	MS35782-2
AA59440/1-002B	MS35782-2
AA59440/1-002C	MS35782-2
AA59440/1-002Z	MS35782-2
AA59440/1-003	MS35782-3
AA59440/1-003B	MS35782-3
AA59440/1-003C	MS35782-3
AA59440/1-003Z	MS35782-3
AA59440/1-004	MS35782-4
AA59440/1-004B	MS35782-4
AA59440/1-004C	MS35782-4
AA59440/1-004Z	MS35782-4
AA59440/1-005	MS35782-5
AA59440/1-005B	MS35782-5
AA59440/1-005C	MS35782-5
AA59440/1-005Z	MS35782-5

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AA59440/1-006	MS35782-6
AA59440/1-006B	MS35782-6
AA59440/1-006C	MS35782-6
AA59440/1-006Z	MS35782-6
AA59440/1-007	MS35782-7
AA59440/1-007B	MS35782-7
AA59440/1-007C	MS35782-7
AA59440/1-007Z	MS35782-7
AA59440/1-008	MS35782-8
AA59440/1-008B	MS35782-8
AA59440/1-008C	MS35782-8
AA59440/1-008Z	MS35782-8

1/ These parts were originally listed in MS35782 as a single part type. Due to special handling/disposal and corrosion resistance requirements the parts have been separated in this CID.

Part number P/N supersession data. These CID specification sheet PINs supersede the following MFR's P/N's as shown in table III. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE III. P/N Supersession data. 1/

Dash number (see figure 1 and figure 2) AA59440/1	MFR's CAGE 2/	MFR's P/N	MFR's CAGE	MFR's P/N	MFR's CAGE	MFR's P/N
001	01276	130	20969	DC100 1/8	93061	DC603-2
001C						
002	01276	140	20969	DC100 1/4	93061	DC603-4
002C						
003	01276	230		N/A	93061	DC603-6
003C				N/A		
004	01276	135		N/A	93061	DC604-2
004C				N/A		
005	01276	145		N/A	93061	DC604-4
005C				N/A		
006	01276	270		N/A	93061	DC604-6
006C				N/A		
007	01276	108		N/A	93061	N/A
007C				N/A		N/A
008	01276	N/A		N/A	93061	N/A
008C		N/A		N/A		N/A

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see marking paragraph of A-A-59440.
2/ Manufacturers products not approved for steam service

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National stock number (NSN). The following is a list of NSN's assigned which correspond to this CID specification sheet. The list in table III is for information only and may not be indicative of all possible NSN's associated with the CID specification sheet. For up to date information on assigned NSN's, please contact the aforementioned DLA Land and Maritime office listed in A-A-59440.

TABLE III. NSN's.

Dash number (see figure 1 and figure 2) AA59440/1	NSN
003	4820-00-174-0339
004	4820-00-752-9040
005	4820-00-849-1220

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

MILITARY INTERESTS

Custodians:
Army – AR
Navy - SH
Air Force – 71
DLA – CC

Review activities:
Army - AT, CR4, GL, MI
Navy – AS, SA

CIVIL AGENCIES AND COORDINATING ACTIVITIES:

GSA – FSS

Preparing activity:

DLA - CC

(Project 4820-2020-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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